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Nondestructively identifying the mechanical behavior of soft tissues using surface deformation with an explicit inverse approach

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A B S T R A C T

Identifying the spatial variation of stiffness properties in soft tissues nondestructively, with limited surface measurements, poses significant challenges. In this paper, we present a novel explicit inverse approach designed to characterize the nonhomogeneous elastic property distribution of soft tissues using only surface displacement datasets. In contrast to the prevalent implicit inverse approach, which focuses on optimizing the elastic properties of individual pixels, our proposed method optimizes the geometric parameters of deformable and movable components, as well as shear moduli of each component. As a result, the proposed approach requires far fewer optimization variables, streamlining the process. Numerical tests conducted in this study demonstrate the superiority of the explicit inverse method over the implicit inverse method, providing much-improved reconstructed results. In particular for a ring structure, while the average relative error of reconstruction using the implicit inverse method can exceed 40 %, the explicit inverse method achieves a remarkable average relative error of only 5 %. Given that surface displacements are easily measurable, the integration of our proposed explicit inverse method with low-cost imaging techniques shows great potential in accurately mapping the elastic property distribution of biological tissues.

Introduction

Biological tissues often display significant heterogeneity. This diversity is evident in various tissues such as skin [1–3], blood vessels [4,5], and corneas [6,7], among others, which exhibit multi-layered structures. These layers offer a variety of functionalities, thus contributing to the complexity of biological systems. Interestingly, the emergence of cancerous tumors introduces an additional level of heterogeneity to these tissues. These tumors typically exhibit greater stiffness compared to their surrounding healthy tissues due to pathological alterations [8]. This stark contrast in mechanical properties can serve as a potential indicator of malignancy. It's also worth noting that the local mechanical properties of soft tissues are not static; they fluctuate in response to various factors such as age

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... [9–11], dietary habits [12], and the presence of pathological conditions [13,14]. Therefore, understanding the nonhomogeneous biomechanical behavior of these tissues is of paramount importance, particularly in a non-destructive manner. Such knowledge can inform medical diagnostics and treatments, potentially leading to better patient outcomes.

Considerable effort has been devoted into research aimed at identifying the nonhomogeneous biomechanical properties of biological tissues. Studies [15–18] have demonstrated the feasibility of estimating the mechanical properties of each layer in multi-layered biological tissues, provided that the thickness of each layer is known beforehand. However, in practice, determining the thickness of each layer prior to addressing the identification problem can be a challenging task. Furthermore, the interface between neighboring layers is seldom flat, a characteristic particularly pronounced in biological tissues. As such, a comprehensive solution requires knowledge of both the topology of the interface between the layers and the biomechanical properties of each layer. Additionally, the detection of tumors embedded within normal tissues introduces another layer of complexity. To accurately diagnose and plan treatment for these conditions, it is crucial to determine not only the elastic modulus values of these tumors but also their topology [19].

A general strategy for tackling this challenge involves identifying the spatial variation of material properties, a process that necessitates solving a large-scale optimization problem to optimize the elastic properties of every pixel in the domain of interest, which can be referred to as the implicit inverse method [20–23]. The implicit inverse method has been successfully employed in tumor detection using real clinical data [8,24]. However, this implicit inverse method has its limitations. It not only demands significant computational resources but is also prone to severe uniqueness issues, which can hamper the reliability of the results. Given that biological tissues are often known a priori to exhibit either layered structures or inclusion-embedded structures, it's feasible to explicitly and simultaneously identify the topology of each nonhomogeneous region and its associated material properties. This approach bypasses some of the computational hurdles of the implicit method. Another inherent drawback of the implicit inverse method lies in the quality of the reconstruction results, which can be compromised if the full-field displacement field cannot be measured with high accuracy. Studies [25,26] that employed the implicit inverse method to reconstruct the nonhomogeneous elastic property distribution of solids using surface deformation have demonstrated this issue. While these methods are capable of visualizing the nonhomogeneous region, the quality of the reconstruction is highly sensitive to noise. This sensitivity arises from a disparity between the large number of unknown optimization variables in the implicit inverse method and the sparsity of the deformation information on the surface. In contrast, the explicit inverse method significantly reduces the total number of optimization variables [27–29], potentially improving the quality of the reconstructed results. This strategy holds promise for enhancing the accuracy of nonhomogeneous biomechanical property identification in biological tissues.

This paper will present an explicit inverse method that utilizes surface deformation measurements to identify the mechanical behavior of two common nonhomogeneous structures: layered structures and inclusion embedded structures. The organization of this work is as follows: In the **Methods** Section, we provide a detailed explanation of the mathematical background underlying the explicit inverse approach. The **Results** Section presents a comparative study between the explicit and implicit inverse approaches, using various numerical examples. We discuss the obtained results in the **Discussion** Section and conclude the paper with a summary in the **Conclusion** Section.

Methods

Forward problem

Given the low elastic modulus of soft tissues, this study will consider finite deformation in the analysis. The strong form of the nonlinear elastic problem is stated as follows:

$$\begin{cases} \mathrm{Div}(\mathbf{P}) = \mathbf{0} & \text{in } \Omega \\ \mathbf{u} = \mathbf{g} & \text{on } \Gamma_g \\ \mathbf{P} \cdot \mathbf{N} = \mathbf{T} & \text{on } \Gamma_T \end{cases} \tag{1}$$

where $\mathbf{P} = \mathbf{F}\mathbf{S}$ is the first Piola–Kirchhoff stress tensor where $\mathbf{F}$ is the deformation gradient and $\mathbf{S}$ is the second Piola–Kirchhoff stress tensor. Ω is the problem domain at the reference configuration. $\mathbf{g}$ is the prescribed displacement on the displacement boundary Γ_g. $\mathbf{T}$ is the prescribed traction on the traction boundary Γ_T. The following restrictions apply to Γ_g and Γ_T: $\Gamma_g \cup \Gamma_T = \partial\Omega$ and $\Gamma_g \cap \Gamma_T = \phi$. We assume the soft tissue of interest obeys the incompressible and neo-Hookean law:

$$\mathbf{S} = \mu\big(\mathbf{I} - \mathbf{C}^{-1}\big) + p\,\mathbf{C}^{-1} \tag{2}$$

where $\mathbf{C} = \mathbf{F}^{\mathsf{T}}\mathbf{F}$ is the Cauchy–Green tensor. p is the hydrostatic pressure.

The weak form of the nonlinear elastic forward problem is stated as: Find $[\mathbf{u}^h, p^h] \in \mathscr{M}^h \times \mathscr{P}^h$ such that

$$\mathscr{A}\Big(\mathbf{w}^h, \mathbf{u}^h, q^h, p^h\Big) \;=\; \int_{\Gamma_T} \mathbf{w}^h \cdot \mathbf{T}\,\mathrm{d}\Gamma, \qquad \forall\,[\mathbf{w}^h, q^h] \in \ell^h \times \mathscr{P}^h \tag{3}$$

$$\mathscr{A}\Big(\mathbf{w}^h, \mathbf{u}^h, q^h, p^h\Big) \;\equiv\; \int_\Omega \nabla\mathbf{w} : \mathbf{P}\,\mathrm{d}\Omega \;+\; \int_\Omega q\,(J-1)\,\mathrm{d}\Omega \;-\; \sum_{i=1}^{NE}\Big(\tau\,\mathrm{div}(\mathbf{F}^h\mathbf{S}^h),\; (\mathbf{F}^h)^{-\mathsf{T}}\nabla q^h\Big)_{\Omega_i} \tag{4}$$

where the superscript h denotes the finite element subspace. $\mathscr{M}^h$, ℓ^h and $\mathscr{P}^h$ are the finite-element subspaces of the function space $\mathscr{M} \equiv \{\mathbf{u} \mid u_i \in H^1(\Omega);\, u_i = g_i \text{ on } \Gamma_g\}$, $\ell \equiv \{\mathbf{w} \mid w_i \in H^1(\Omega);\, w_i = 0 \text{ on } \Gamma_g\}$ and $\mathscr{P} \subseteq L^2(\Omega)$, respectively. The third term is the stabilized term to address the volumetric locking induced by the incompressibility where NE is the total number of elements in the discretized domain. $\tau = \dfrac{\alpha h^2}{2\mu}$ represents the stabilization factor, where h is the characteristic element length. α is set to 1.0 [30].

Topological description of nonhomogeneous elastic property distribution

To describe the topology of the nonhomogeneous elastic property distribution, we introduce the B-spline function to express a curve:

$$\mathbf{C}(t) = \sum_{i=0}^{n+1} N_{i,p}(t)\, \mathbf{P}_i, \qquad (a \leq t \leq b) \tag{5}$$

where $N_{i,p}(t)$ is the p-th order B-spline basis function generated by a knot vector $\{t_0, t_1, \ldots, t_{n+2+p}\}$. $\mathbf{P}_i\,(i = 0, \ldots, n+1)$ is the i-th control point.

The function $N_{i,p}(t)$ can be determined by Eq. (6):

$$\begin{cases} N_{i,0}(t) = \begin{cases} 1, & \text{if } t_i \leq t \leq t_{i+1} \\ 0, & \text{otherwise} \end{cases} & (i = 0, \ldots, n+1;\; p = 0) \\ N_{i,p}(t) = \dfrac{t - t_i}{t_{i+p} - t_i} N_{i,p-1}(t) + \dfrac{t_{i+p+1} - t}{t_{i+p+1} - t_{i+1}} N_{i+1,p-1}(t), & (i = 0, \ldots, n+1;\; p \geq 1) \end{cases} \tag{6}$$

To capture various types of nonhomogeneous elastic property distributions, we employ B-splines to approximate both open and closed curves, as depicted in Fig. 1. The open curve is utilized to represent the interface shape between adjacent layers in a layered composite, while the closed curve represents the shape of inclusions embedded within a soft solid. To avoid the self-intersection of the closed B-spline curve, we define the control points of the closed B-spline curve in Eq. (7):

$$\mathbf{P}_i = \begin{cases} \left(x_o + d_i \sin\left(i\theta - \dfrac{\theta}{2}\right),\; y_o + d_i \cos\left(i\theta - \dfrac{\theta}{2}\right)\right), & \left(\theta = \dfrac{2\pi}{n};\; i = 1, \ldots, n\right) \\ \mathbf{P}_0 = \mathbf{P}_{n+1} = \dfrac{\mathbf{P}_1 + \mathbf{P}_n}{2} \end{cases} \tag{7}$$

where (x_o, y_o) is the location of the center of the closed B-spline curve, d_i is the distance between the i-th control point and the center of the closed curve.

With the curve expressed by the B-spline function, we can define each nonhomogeneous region in the problem domain by

[Figure 1 — placeholder]

(a) Open curve: eight control points $\mathbf{P}_0, \mathbf{P}_1, \ldots, \mathbf{P}_7$ connected by dashed blue segments, with a smooth orange B-spline curve passing through the polygon.

(b) Closed curve: nine control points $\mathbf{P}_0 = \mathbf{P}_9, \mathbf{P}_1, \ldots, \mathbf{P}_8$ distributed around a center (x_o, y_o), with radial distances $d_1, d_2, \ldots, d_8$ and angular spacing $\theta = 2\pi/n$. The closed B-spline curve (orange) is inscribed in the dashed control polygon, with $\mathbf{P}_0 = \mathbf{P}_9 = (\mathbf{P}_1 + \mathbf{P}_8)/2$.

Fig. 1. Two examples of the curve expressed by B-spline functions.

introducing the following topological description function (TDF):

$$\begin{cases} \phi^i(\mathbf{x}) < 0 & \text{if } \mathbf{x} \in \Omega^i \\ \phi^i(\mathbf{x}) = 0 & \text{if } \mathbf{x} \in \partial\Omega^i \\ \phi^i(\mathbf{x}) > 0 & \text{if } \mathbf{x} \in \Omega \setminus \Omega^i \end{cases} \tag{8}$$

For the layered structure, ϕ^i is the TDF of the i-th interface and Ω^i is the region occupied from the curve C^i and C^0. The mathematical expression of TDF for a layered structure can be defined as:

$$\phi^i(\mathbf{x}) = r(t) - r^i(t) \tag{9}$$

where $r^i(t)$ is the distance between the curve C^i and C^0 at any arbitrary point on the domain. To this end, the m-th layer of the structure ($m > 1$) can be defined by $\phi^{m-1}(\mathbf{x}) > 0$ and $\phi^m(\mathbf{x}) \leq 0$, except for the first layer $\phi^1(\mathbf{x}) < 0$ and last layer $\phi^n(\mathbf{x}) > 0$. To this end, A bilayer structure can be expressed by the proposed method as shown in Fig. 2. Note that we assume that each layer is homogeneous and did not consider spatial variations within a layer.

For the domain with inclusion embedded, ϕ^i is the TDF of the i-th component and Ω^i is the region occupied by the i-th component. For an inclusion embedded composite, the mathematical expression of TDF is:

$$\phi^i(\mathbf{x}) = \sqrt{\left(x - x_o^i\right)^2 + \left(y - y_o^i\right)^2} - r^i\left(\theta^i\right) \tag{10}$$

where θ^i is the angle from the point (x, y) to the center of the i-th component $\left(x_o^i, y_o^i\right)$ in polar coordinate system. $r^i(\theta^i)$ is the distance from the point with the angle of θ^i on $\partial\Omega^i$ to the center $\left(x_o^i, y_o^i\right)$ as shown in Fig. 3.

With the defined TDF, the spatial variation of the shear modulus of a layered composite can be defined based on the Boolean operators [31]:

$$\mu(\mathbf{x}) = \mu^0\big(1 - H(\phi^1(\mathbf{x}))\big) + \sum_{i=1}^{n-1}\big[\mu^i\, H(\phi^i(\mathbf{x}))\big(1 - H(\phi^{i+1}(\mathbf{x}))\big)\big] + \mu^n\, H(\phi^n(\mathbf{x})) \tag{11}$$

where μ^0 is the shear modulus of background. μ^n is the shear modulus of the last layer. $H(z)$ is the Heaviside function. If there is no interface curve C^i, $\mu(\mathbf{x}) = \mu^0$ is a homogeneous domain. However, the TDF definitions of composite with embedded inclusions are different, that is:

$$\boldsymbol{\mu}(\boldsymbol{x}) = \boldsymbol{\mu}^0 \prod_{i=1}^{n} \mathbf{H}\big(\boldsymbol{\phi}^i(\boldsymbol{x})\big) + \sum_{i=1}^{n}\Big(\boldsymbol{\mu}^i\big(1 - \mathbf{H}(\boldsymbol{\phi}^i(\boldsymbol{x}))\big)\Big) \tag{12}$$

As the Heaviside function is a discontinuous function, we will utilize the following function to approximate the Heaviside function so that we can evaluate the spatial gradient of the Heaviside function:

$$H(z) = 0.5 \times \left(1 + \tanh\left(\frac{z}{\varepsilon}\right)\right) \tag{13}$$

where ε is a small constant. With the decrease of ε, the approximation of the Heaviside function is closer to the corresponding analytical expression (see Fig. 4). However, a smaller ε might lead to an extremely large gradient at z approaching to zero. Thus, the small constant ε should be carefully chosen.

Inverse problem

After discussing the TDF, the inverse problem can be stated as: Given n_{meas} surface measured displacement fields, find the optimal geometric vectors $\mathbf{r}^1, \mathbf{r}^2, \ldots, \mathbf{r}^m$ and the shear moduli $\mu^0, \mu^1, \ldots, \mu^m$, such that the following objective function is minimized:

$$\Pi = \frac{1}{2} \sum_{i_{\mathrm{meas}}=1}^{n_{\mathrm{meas}}} \int \left|\mathbf{u}_i^{\mathrm{com}} - \mathbf{u}_i^{\mathrm{meas}}\right|^2 \, d\partial\Omega_{i_{\mathrm{meas}}} + \mathrm{Reg}(\mu) \tag{14}$$

[Figure 2 — placeholder]

Bilayer structure illustrating the TDF construction. The bottom layer μ^0 (teal) is bounded below by the reference curve C^0; the top layer μ^1 (pink) is bounded above by the interface curve C^1. Six control points $\mathbf{P}_1^1, \mathbf{P}_2^1, \ldots, \mathbf{P}_6^1$ define the B-spline interface C^1 (orange). Radial distances $r_1^1, r_2^1, \ldots, r_6^1$ connect each control point back to C^0, parameterised along t.

Fig. 2. An example of the nonhomogeneous shear modulus distribution of a bilayer structure.

[Figure 3 — placeholder]

Square light-teal domain with background shear modulus μ^0, containing an irregular pink inclusion region with shear modulus μ^1 and boundary $\partial\Omega^1$. Eight control points $\mathbf{P}_1^1, \mathbf{P}_2^1, \ldots, \mathbf{P}_8^1$ are distributed around the boundary, connected to each other by dashed blue segments (the control polygon) and smoothly interpolated by an orange closed B-spline curve. Eight radial distances $d_1^1, d_2^1, \ldots, d_8^1$ connect the inclusion center (x_0^1, y_0^1) to each control point, illustrating the polar parameterization used in Eq. (7) and Eq. (10).

Fig. 3. The nonhomogeneous shear modulus distribution of an inclusion embedded composite.

[Figure 4 — placeholder]

Graph of $H(z) = 0.5 \times (1 + \tanh(z/\varepsilon))$ (the smoothed Heaviside from Eq. (13)) plotted over $z \in [-0.5, 0.5]$, with $H(z) \in [0, 1]$ on the vertical axis. Four curves show the effect of decreasing ε:

$\varepsilon = 0.01$ sharpest transition (nearly a true step)
$\varepsilon = 0.05$ moderate transition
$\varepsilon = 0.10$ smoother transition
$\varepsilon = 0.15$ smoothest transition

All four curves pass through $(0, 0.5)$. As $\varepsilon \to 0$ the approximation converges to the true Heaviside; as ε grows the transition band widens.

Fig. 4. An approximation of Heaviside function.

$$\mathrm{Reg}(\mu) = \frac{1}{2}\,\alpha \int_{\Omega} \sqrt{|\nabla \mu|^2 + c^2}\, d\Omega \tag{15}$$

For the layered structure, $\mathbf{r}^i = \left[r_1^i(t_1),\, r_2^i(t_2),\, \cdots,\, r_n^i(t_n)\right]$ is the geometric parameter vector representing the distance between the i-th interface and C^0 at the location t_k. For the structure with inclusions embedded, the geometric parameter vector $\mathbf{r}^i = \left[r_1^i,\, r_2^i,\, \cdots,\, r_n^i\right] = \left[x_0^i,\, y_0^i,\, d_1^i,\, \cdots,\, d_{n-2}^i\right]$ is consisting of the coordinates of the center and distances between the control points to the center of i-th component. In Eq. (14), the discrepancy between the computed displacement field and measured displacement field on partial surface $\partial\Omega_{i_{\text{meas}}}$ is minimized in L-2 norm. The computed displacement field is obtained by solving a forward problem using the finite element method [32]. Due to the influence of noise in measurements, we introduce a total variation diminishing regularization (TVD) term (the second term in Eq. (14)), and the specific form is given by Eq. (15). α is the regularization factor which controls the significance of the regularization on the objective function. c is a small constant to avoid singularities when evaluating the gradient of the regularization term with respect to optimization variables. It should be noted that there exist alternative forms for the regularization term apart from the domain integral forms. Nevertheless, we have chosen to adhere to the use of domain integrals due to practical considerations. In particular, the explicit inverse solver has been developed based on the established framework of the existing implicit inverse solver. This continuity in methodology ensures consistency and facilitates seamless integration between the two solvers [33]. The specific value of the regularization factor is chosen based on the smoothness criteria [22].

The optimization problem is solved by the limited Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) method which requires the objective function and the gradients of the objective function with respect to the optimization variables. The gradients of the objective function with respect to the optimization variables are obtained by the adjoint method which has been thoroughly discussed in [27]. The flow chart of the procedure to solve the inverse problem is shown in Fig. 5.

To test the feasibility of the proposed inverse scheme, we adopt the displacement datasets generated by a forward finite element simulation with a target shear modulus distribution. The white Gaussian random is also added into the simulated surface displacement field. The noise level is defined by $\sqrt{\sum_{k=1}^{N_{\text{node}}} \left(u_k - u_k^{\text{noise}}\right)^2 \Big/ \sum_{k=1}^{N_{\text{node}}} (u_k)^2} \times 100\,\%$, where u_k and u_k^{noise} are the simulated and noisy displacement of k-th node, N_{node} is the total number of nodes in interest domain. In comparison, we also solve inverse problems using the prevalent implicit inverse method based on the adjoint method where the shear moduli are nodally defined [22]. To quantitatively analyze the performance of the proposed method, we define the average relative error by $\frac{1}{N_{\text{node}}} \sum_{k=1}^{N_{\text{node}}} \left| \left(\mu_k - \mu_k^{\text{target}}\right) / \mu_k^{\text{target}} \right| \times 100\,\%$, where μ_k is the recovered shear modulus value of k-th node and μ_k^{target} is the target shear modulus value of k-th node.

Gradient computation

To efficiently compute the gradient of the objective function with respect to the optimization variable, we have adopted the adjoint method, which involves solving the associated adjoint equation. The adjoint equation is given by [34,35]:

$$\mathscr{B}\Big(\mathbf{w}^h, q^h, \delta\mathbf{u}^h, \delta p^h;\, \mathbf{u}^h, p^h\Big) = -\big(\delta\mathbf{u}^h,\, \mathbf{u}^h - \mathbf{u}^m\big)_{\partial\Omega}, \qquad \forall\, [\delta\mathbf{u}^h, \delta p^h] \in \mathscr{M}^h \times \mathscr{P}^h \tag{16}$$

where $\mathscr{B}(\mathbf{w}^h, q^h, \delta\mathbf{u}^h, \delta p^h;\, \mathbf{u}^h, p^h) = \frac{d}{d\varepsilon}\mathscr{A}^*(\mathbf{w}^h, q^h, \mathbf{u}^h + \varepsilon\,\delta\mathbf{u}^h, p^h + \varepsilon\,\delta p^h)\big|_{\varepsilon=0}$ is the linearization of $\mathscr{A}^*$. With the acquired state variables $\mathbf{u}^h, p^h$, we can solve for the dual variables $\mathbf{w}^h, q^h$ using Eq. (16). To this end, the gradient of the objective function with respect to the shear modulus:

$$\delta\Pi = D_\mu \Pi\, \delta\mu = \mathscr{C} + \alpha \int_\Omega \frac{\nabla\mu \cdot \nabla\delta\mu}{\sqrt{|\nabla\mu|^2 + c^2}}\, d\Omega \tag{17}$$

$\mathscr{C} = \frac{d}{d\varepsilon}\mathscr{A}^*(\mathbf{w}^h, q^h, \mathbf{u}^h, p^h, \mu + \varepsilon\,\delta\mu)\big|_{\varepsilon=0}$. This can be used to efficiently compute the gradient of the objective function with respect to

[Figure 5 — placeholder: optimization flowchart]

$\boxed{\text{Give initial guess: } \mathbf{r}^1, \mathbf{r}^2, \ldots, \mathbf{r}^m \text{ and } \mu^0, \mu^1, \ldots, \mu^m}$

$\downarrow$ *Finite Element Method*

$\boxed{\text{Solve forward problem} \rightarrow \text{computed displacement } \mathbf{u}^{\text{com}}}$

$\downarrow$ *Adjoint Method*

$\boxed{\text{Evaluate } \Pi \text{ and its gradients w.r.t. optimization variables}}$

$\downarrow$

$\boxed{\text{Is } \Pi \text{ minimized?}} \xrightarrow{\textit{No}} \boxed{\text{Update variables } (\textit{L-BFGS})} \rightarrow$ loop back $\uparrow$

$\downarrow$ *Yes*

$\boxed{\text{Stop and output reconstruction results}}$

Fig. 5. The flow chart of the proposed explicit inverse method.

nodal or elemental shear moduli. If the shear modulus is nodally defined, we can use the same interpolation function with the displacement field. In this case, the shear modulus can be expressed as:

$$\mu \approx \mu^h = \sum_{i=1}^{N} N_i \, \mu_i \tag{18}$$

where N is the total number of nodes in the finite element model and N_i is the finite element shape function. μ_i is the nodal shear modulus.

In the explicit inverse method, since shear modulus distribution is a function of geometric parameters and the shear modulus in each region, the gradient of the objective function with respect to the optimization variables can be calculated using the chain rule. To this end, we should seek $\delta\mu$ corresponding to a change in any optimization variable from β to $\beta + \delta\beta$:

$$\delta\mu = \frac{\partial\mu}{\partial\beta}\,\delta\beta \tag{19}$$

Thereby, comparing to the implicit inverse method, we need to evaluate the $\partial\mu/\partial\beta$.

(a) Layered structure

The evaluation of the gradient of geometric optimization variables in a layered structure with $n+1$ layers should be discussed considering three cases. For the m-th geometric optimization variable r_m^1 at the first interface C^1, $\frac{\partial\mu}{\partial r_m^1}$ is given by:

$$\frac{\partial\mu}{\partial r_m^1} = \left[-\mu^0 + \mu^1\big(1 - H\big(\phi^2(\mathbf{x})\big)\big)\right] \frac{\partial H\big(\phi^1(\mathbf{x})\big)}{\partial\phi^1(\mathbf{x})} \frac{\partial\phi^1(\mathbf{x})}{\partial r_m^1} \tag{20}$$

For the m-th geometric optimization variable r_m^n at the last interface C^n, $\frac{\partial\mu}{\partial r_m^n}$ is given by:

$$\frac{\partial\mu}{\partial r_m^n} = \left(-\mu^{n-1} H\big(\phi^{n-1}(\mathbf{x})\big) + \mu^n\right) \frac{\partial H\big(\phi^n(\mathbf{x})\big)}{\partial\phi^n(\mathbf{x})} \frac{\partial\phi^n(\mathbf{x})}{\partial r_m^n} \tag{21}$$

For the m-th geometric optimization variable at the intermediate interface C^i $(2 \leq i \leq (n-1))$, $\frac{\partial\mu}{\partial r_m^i}$ is given by:

$$\frac{\partial\mu}{\partial r_m^i} = \left[-\mu^{i-1} H\big(\phi^{i-1}(\mathbf{x})\big) + \mu^i\big(1 - H\big(\phi^{i+1}(\mathbf{x})\big)\big)\right] \frac{\partial H\big(\phi^i(\mathbf{x})\big)}{\partial\phi^i(\mathbf{x})} \frac{\partial\phi^i(\mathbf{x})}{\partial r_m^i} \tag{22}$$

In Eqs. (20–22), $\frac{\partial H(\phi^j(\mathbf{x}))}{\partial\phi^j(\mathbf{x})}$ $(1 \leq j \leq n)$ can be evaluated by Eq. (13) as the Heaviside function is approximated by a continuous and differentiable function. $\frac{\partial\phi^j(\mathbf{x})}{\partial r_m^j}$ can be evaluated by the finite difference method.

For the shear modulus of the background μ^0, $\frac{\partial\mu}{\partial\mu^0}$ is given by:

$$\frac{\partial\mu}{\partial\mu^0} = 1 - H\big(\phi^1(\mathbf{x})\big) \tag{23}$$

For the shear modulus of the last layer μ^n, we have

$$\frac{\partial\mu}{\partial\mu^n} = H(\phi^n(\mathbf{x})) \tag{24}$$

For the shear modulus of the layer μ^i $(1 \leq i \leq (n-1))$, we have

$$\frac{\partial \mu}{\partial \mu^i} = H\big(\phi^i(\mathbf{x})\big)\left(1 - H\big(\phi^{i+1}(\mathbf{x})\big)\right) \tag{25}$$

(b) Domain with inclusion embedded

For each m-th geometric optimization variables of j-th component r^j_m, $\frac{\partial \mu}{\partial r^j_m}$ is given by:

$$\frac{\partial \mu}{\partial r^j_m} = \left[\mu^0 \prod_{\substack{i=1 \\ i \neq j}}^{n} H\big(\phi^i(\mathbf{x})\big) - \mu^j\right] \frac{\partial H\big(\phi^j(\mathbf{x})\big)}{\partial \phi^j(\mathbf{x})} \frac{\partial \phi^j(\mathbf{x})}{\partial r^j_m} \tag{26}$$

For the shear modulus of the background μ^0, we have:

$$\frac{\partial \mu}{\partial \mu^0} = \prod_{i=1}^{n} H\big(\phi^i(\mathbf{x})\big) \tag{27}$$

For the shear modulus μ^j of the j-th component, $\frac{\partial \mu}{\partial \mu^j}$ is given by:

$$\frac{\partial \mu}{\partial \mu^j} = 1 - H\big(\phi^j(\mathbf{x})\big) \tag{28}$$

Results

Case 1: A layered rectangular structure

In the first numerical example, we analyze a layered rectangular structure, illustrated in Fig. 6. The dimensions of this structure are 1.0 mm × 0.2 mm. To discretize the problem domain, we utilize 7200 linear triangular elements. In the forward finite element simulation, a concentrated force is applied to the top edge, while the bottom edge was restricted the vertical motion and the center point is fully fixed. In the inverse problem, our focus lies in utilizing displacement datasets collected from the top edge as inputs for the inverse scheme. Acknowledging that relying on a single surface displacement dataset may not produce optimal reconstruction results, we change the location of the concentrated force and incorporate multiple surface displacement datasets to solve the inverse problem. The target shear modulus distributions for one (the homogenous case), two, and three layers are depicted in Fig. 6(a), (b), and (c), respectively. The shear modulus values of each layer in the structure are of a similar order to that of skin tissue [36]. Considering the uncertainty regarding the total number of layers, we initially assume a configuration with six layers for the inverse problem, as depicted in Fig. 7(a). For the explicit inverse scheme, we utilize 10 control points to interpolate each interface. Thus, the total number of optimization variables is 56 including 50 geometric variables and 6 shear moduli.

From Fig. 7, it can be seen that both the shape of the interfaces between the neighboring layers and the shear modulus values of each layer varies and finally reach to the target shear modulus distribution (Sample 1). Note that it takes only 52 iterations for convergence. Figs. 8–10 are the reconstructed shear modulus distributions for Samples 1–3 using explicit and implicit inverse approaches for the noise free cases, respectively. In the implicit inverse method, the total number of optimization variables is linearly proportional to the mesh size. In this case, the total number of optimization variables is 3691, which is equal to the total number of nodes. The reconstructed results demonstrate that both inverse methods can yield decent reconstruction with noise free cases. Further, the implicit method fails to recover the heterogeneity at the upper corner compared with the results by the explicit inverse method. Since the total number of optimization variables is remarkably reduced in the explicit inverse method, the total number of minimization iterations is reduced. The total number of minimization iterations for the explicit inverse method is approximately 10 times less than that required for the implicit inverse method. Furthermore, the shear modulus reconstruction obtained through the implicit

[Figure 6 — placeholder]

Three layered rectangular structures (1.0 mm × 0.2 mm), each showing the target shear modulus (SM) distribution in MPa with a jet-style colorbar on the right.

(a) Sample 1: homogeneous, single layer, SM = 3.00 MPa (uniform blue). Downward arrows on top edge indicate concentrated force locations. Red text "displacement measurement" with leftward arrows on left side.

(b) Sample 2: two layers. Bottom layer SM $\approx$ 3 MPa (blue), top layer SM $\approx$ 10 MPa (red). Green circles on top boundary, red circles on bottom boundary mark displacement measurement points.

(c) Sample 3: three layers (blue–red–blue sandwich). SM ranges 1.00–10.00 MPa. Same boundary marker pattern as (b).

All three samples share the same mesh (7200 triangular elements) and boundary conditions (top-edge force, bottom-edge vertical constraint, center point fixed).

Fig. 6. The target shear modulus distribution of three samples with different number of layers (Unit: MPa). SM stands for shear modulus.

[Figure 7 — placeholder]

Six-panel grid showing the optimization trajectory for the explicit inverse scheme on **Sample 1** (the homogeneous case, target SM = 3.00 MPa), starting from a 6-layer initial guess. Each panel is a rectangular shear-modulus distribution (1.0 mm ×0.2 mm) with a jet colorbar.

(a) Initial guess	uniform blue, SM = 1.00 MPa everywhere
(b) Iteration 5	SM range 1.27–1.95, layered artifacts emerging
(c) Iteration 12	SM range 1.50–3.11, layered structure developed
(d) Iteration 24	SM range 1.63–3.03, approaching target
(e) Iteration 48	SM range 2.97–3.00, nearly converged
(f) Iteration 52 (Final)	uniform red, SM = 3.00 MPa everywhere

The trajectory shows the 6-layer parameterisation collapsing to a uniform solution over 52 iterations — because the target is homogeneous, the optimiser correctly drives all six layers to the same shear modulus.

Fig. 7. The optimization process with six layers initially utilizing the explicit inverse scheme for Sample 1 (Without noise, Unit: MPa).

[Figure 8 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on Sample 1 (homogeneous, target SM = 3.00 MPa), with 3, 6, and 9 surface displacement datasets as inputs.

Explicit (left column): all three rows show a perfectly uniform shear-modulus field at SM = 3.00 MPa (uniform blue, colorbar pinned 3.00–3.00). The explicit parameterisation recovers the homogeneous target exactly.

Implicit (right column): all three rows show tiny spatial artifacts around the true value of 3.00 MPa — (d) 3 fields: SM range 2.9995–3.0001; (e) 6 fields: SM range 2.9992–3.0001; (f) 9 fields: SM range 3.0000–3.0002. The color blobs are visible even though the numerical error is on the order of 10^{-4}–10^{-3} MPa.

Takeaway: the explicit method produces a perfectly uniform field on the homogeneous case; the implicit method is quantitatively accurate but shows residual spatial noise patterns that decrease (slightly) as more displacement datasets are incorporated.

Fig. 8. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 1 (Without noise, Unit: MPa).

inverse method shows a slight spatial variation, which can be enhanced by incorporating more surface datasets.

After adding $1\,\%$ noise to the surface displacement datasets, we observed that the shear modulus distributions were accurately mapped for both inverse methods in the homogeneous case (refer to Fig. 11). The average relative error for both implicit and explicit inverse methods was less than $0.2\,\%$ (see Table 1). However, when it came to Sample 2 and 3 (see Fig. 12, Fig. 13), the interface between neighboring layers in the implicit inverse method's results was not as clear as the target. As the total number of layers increases, the challenges in solving the inverse problems escalate accordingly. Consequently, the relative error for Sample 2 and Sample 3 exhibits a lower quality compared to Sample 1 (see Table 1). Besides, the average relative error for the implicit inverse method was

[Figure 9 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on **Sample 2** (the two-layer target from Fig. 6b, nominal SM values $\mu^0 \approx 3\,\text{MPa}$, $\mu^1 \approx 10\,\text{MPa}$), using 3, 6, and 9 surface displacement datasets as inputs.

Explicit (left column): bilayer structure recovered with a clean interface. Colorbar ranges: (a) 3 fields, SM $\in [1.00, 11.49]$ MPa; (b) 6 fields, SM $\in [1.00, 11.63]$ MPa; (c) 9 fields, SM $\in [1.00, 11.47]$ MPa.

Implicit (right column): bilayer structure recovered but with a visibly rougher interface between the two layers; more artifacts at the top edge. Colorbar ranges: (d) 3 fields, SM $\in [1.00, 10.95]$ MPa; (e) 6 fields, SM $\in [1.00, 10.77]$ MPa; (f) 9 fields, SM $\in [1.00, 11.13]$ MPa.

Takeaway: both methods recover the two-layer structure; the explicit method produces a cleaner interface, while the implicit method's interface has more small-scale roughness. Slight overshoot in the top-layer SM values (colorbar max above the nominal 10 MPa) is visible in all six panels.

Fig. 9. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 2 (Without noise, Unit: MPa).

[Figure 10 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on **Sample 3** (the three-layer target from Fig. 6c: blue bottom layer $\mu^0 \approx 3\,\text{MPa}$, green middle layer $\mu^1 \approx 6\,\text{MPa}$, red top layer $\mu^2 \approx 10\,\text{MPa}$), using 3, 6, and 9 surface displacement datasets as inputs.

Explicit (left column): three-layer stacked structure recovered, with the middle (green) layer visibly distinct from the top (red) and bottom (blue) layers. Colorbar ranges: (a) 3 fields, SM $\in [1.00, 10.49]$ MPa; (b) 6 fields, SM $\in [1.00, 10.35]$ MPa; (c) 9 fields, SM $\in [1.00, 10.44]$ MPa.

Implicit (right column): three-layer structure recovered but with fuzzier interfaces between adjacent layers. Colorbar ranges: (d) 3 fields, SM $\in [1.00, 10.11]$ MPa; (e) 6 fields, SM $\in [1.00, 10.26]$ MPa; (f) 9 fields, SM $\in [1.00, 10.19]$ MPa.

Takeaway: as layer count increases ($1 \to 2 \to 3$), the quality gap between explicit and implicit widens. Sample 3's three-layer structure is harder for the implicit method to resolve cleanly than Sample 2's two-layer structure. More displacement datasets ($3 \to 6 \to 9$) improve both methods but do not close the gap.

Fig. 10. The reconstructed results with 3, 6 and 9 surface displacement datasets corresponding to Sample 3 (Without noise, Unit: MPa).

slightly larger than that for the explicit inverse method (see Table 1). Furthermore, it's worth noting that the quality of the reconstruction achieved by both inverse methods improved with an increase in the total number of measurements. By utilizing 9 boundary displacement measurements, the average relative error could be reduced to less than 10 %. However, it should be mentioned that each layer was not accurately mapped as the homogeneous case when using the implicit inverse method.

Additionally, we increased the noise level to 5 % and observed that the reconstructed results obtained by the explicit inverse method still outperform those obtained by the implicit inverse method (see Fig. 14, Fig. 15, Fig. 16). For the explicit inverse method, the resulting relative error in the case with 5 % noise is generally larger than that in the case with 1 % noise (see Table 1). However, even with a 5 % noise level, the relative error for the explicit inverse method remains lower than that for the implicit inverse method

[Figure 11 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on **Sample 1** (homogeneous, target SM = 3.00 MPa), with **1 % noise** added to the measured surface displacements. Three rows correspond to 3, 6, and 9 displacement datasets.

Explicit (left column): (a) 3 fields: SM $\in [3.0013, 3.0013]$ MPa, uniform red; (b) 6 fields: SM $\in [3.0000, 3.0001]$ MPa, visible green/yellow patches emerge (non-monotone: noisier than (a) or (c)); (c) 9 fields: SM $\in [2.9997, 2.9997]$ MPa, uniform red.

Implicit (right column): (d) 3 fields: SM $\in [2.9981, 3.0083]$ MPa, visible blue patch along the left edge; (e) 6 fields: SM $\in [2.9978, 3.0006]$ MPa, small central blue blob; (f) 9 fields: SM $\in [2.9995, 2.9998]$ MPa, small blue patches remain near the edges.

Takeaway: under 1 % noise, both methods stay quantitatively close to the true 3.00 MPa value (≤ 0.01 MPa deviation). The explicit method's 6-field result is the noisiest explicit panel on this page — an artifact not discussed in the surrounding prose.

Fig. 11. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 1 (1 % noise is introduced into the measured data, Unit: MPa).

Table 1

Comparison of the average relative error in the shear modulus distribution obtained by implicit inverse and explicit inverse methods for the layered structure.

Inverse method		Explicit inverse method			Implicit inverse method		
Number of surface displacement datasets		3	6	9	3	6	9
1 % noise	Sample 1	0.04 %	0.00 %	0.01 %	0.18 %	0.01 %	0.01 %
	Sample 2	11.56 %	6.40 %	4.53 %	13.20 %	7.44 %	4.69 %
	Sample 3	14.03 %	6.84 %	6.71 %	14.55 %	11.37 %	8.58 %
5 % noise	Sample 1	0.36 %	0.22 %	0.003 %	0.11 %	0.15 %	0.003 %
	Sample 2	15.29 %	11.24 %	8.12 %	24.44 %	11.96 %	8.61 %
	Sample 3	25.00 %	9.20 %	8.14 %	19.03 %	13.61 %	10.58 %

with a 1 % noise level. Furthermore, we provided the values of the regularization factors for each case depicted in Figs. 8–16 in Table 2.

Case 2: A layered ring structure

In the second numerical example, we focused on a layered ring structure, as illustrated in Fig. 17. To discretize the problem domain, we employed 7610 triangular elements. The deformation of the ring structure was induced by applying inner pressures ranging from 10 mmHg to 50 mmHg, and we were only able to "measure" the displacement on the inner wall. The shear modulus values of the inner and outer layers were 0.129 MPa and 0.386 MPa [37], respectively. Our approach for the inverse problem relied on utilizing the measured displacement on the inner wall to determine the shear modulus distribution. Initially, we provided an initial guess of the shear modulus distribution, as shown in Fig. 18(a), where four layers were predefined. The shear modulus values, from the outermost layer to the innermost layer, ranged from 0.01 MPa to 0.04 MPa. During the inverse problem-solving process, we optimized both the shape of the interface between neighboring layers and the shear modulus value of each layer. In the initial assumption, the ring structure is considered to consist of 4 layers. For the explicit inverse method, each interface is represented by 14 control points, leading to a total of 56 optimization variables, which includes 4 shear moduli. On the other hand, the implicit inverse method involves a much larger number of optimization variables, specifically 4030 variables in total. Fig. 18 displays the variation of the reconstructed shear modulus distribution using the explicit inverse method for the noise free case. Notably, the reconstruction achieved through the explicit method closely resembled the ground truth. However, the same cannot be said for the implicit inverse method, where the reconstructed shear modulus distribution performed significantly worse than the explicit inverse method even for the noise free data (see Fig. 19(d)-(f)).

[Figure 12 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on **Sample 2** (the two-layer target: background $\mu^0 \approx 3\,\text{MPa}$, top layer $\mu^1 \approx 10\,\text{MPa}$), with **1 % noise** added to the measured surface displacements, using 3, 6, and 9 displacement datasets.

Explicit (left column): bilayer structure recovered with a relatively clean (though slightly wavy) interface. Colorbar ranges: (a) 3 fields, SM $\in [1.00, 11.33]\,\text{MPa}$; (b) 6 fields, SM $\in [1.00, 11.43]\,\text{MPa}$; (c) 9 fields, SM $\in [1.00, 11.34]\,\text{MPa}$.

Implicit (right column): bilayer structure visible but with significant artifacts, especially in panel (d) where a large green/blue patch intrudes into the red top layer from the left edge. Colorbar ranges: (d) 3 fields, SM $\in [1.00, 14.60]\,\text{MPa}$ — a **46 % overshoot** above the nominal 10 MPa top layer, matching the paper's claim that the implicit method's average relative error "could exceed 40 %" under 1 % noise; (e) 6 fields, SM $\in [1.00, 10.60]\,\text{MPa}$; (f) 9 fields, SM $\in [1.00, 10.71]\,\text{MPa}$.

Takeaway: the noise amplifies the implicit method's difficulty with interface recovery dramatically in the 3-field case; 6 and 9 fields bring it back toward the nominal value but with visibly rougher interfaces than the explicit method.

Fig. 12. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 2 (1 % noise is introduced into the measured data, Unit: MPa).

[Figure 13 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on **Sample 3** (the three-layer target: blue bottom layer $\mu^0 \approx 3\,\text{MPa}$, green middle layer $\mu^1 \approx 6\,\text{MPa}$, red top layer $\mu^2 \approx 10\,\text{MPa}$), with **1 % noise** added to the measured surface displacements, using 3, 6, and 9 displacement datasets.

Explicit (left column): three-layer structure recovered with wavy but distinct interfaces between the layers. Colorbar ranges: (a) 3 fields, SM $\in [1.01, 10.63]\,\text{MPa}$; (b) 6 fields, SM $\in [1.00, 10.25]\,\text{MPa}$; (c) 9 fields, SM $\in [1.00, 10.38]\,\text{MPa}$.

Implicit (right column): three-layer structure noticeably noisier, with fuzzier interfaces and small artifact patches. Colorbar ranges: (d) 3 fields, SM $\in [1.00, 11.78]\,\text{MPa}$; (e) 6 fields, SM $\in [1.00, 11.58]\,\text{MPa}$; (f) 9 fields, SM $\in [1.00, 9.96]\,\text{MPa}$ — notably, the 9-field implicit result **undershoots** the nominal 10 MPa target, unusual relative to most other panels in the paper which tend to overshoot.

Takeaway: as with Sample 2 under noise, the explicit method produces cleaner interfaces; but the quantitative colorbar-max values on Sample 3 are closer between explicit and implicit than on Sample 2 because the three-layer structure constrains both methods more tightly than the two-layer structure.

Fig. 13. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 3 (1 % noise is introduced into the measured data, Unit: MPa).

When 1 % noise was introduced, the explicit inverse method continued to perform admirably, producing high-quality reconstructed results (see Fig. 20). However, the performance of the implicit inverse method continued to be inferior to that of the explicit inverse method. The average relative error for the implicit inverse method could exceed 40 %, indicating substantial inaccuracies in the reconstructed results. On the other hand, the average relative error for the explicit inverse method remained relatively low, around 5 %, signifying its superior ability to handle the noise and produce more accurate reconstructions (see Table 3). The values of regularization factors used in each case from Fig. 19 and Fig. 20 are listed in Table 4.

[Figure 14 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on **Sample 1** (the homogeneous target, true SM = 3.00 MPa), with **5 % noise** added to the measured surface displacements, using 3, 6, and 9 displacement datasets.

Explicit (left column): the higher noise level corrupts the homogeneous recovery. Colorbar ranges: (a) 3 fields, SM $\in [2.9641, 3.0096]$ MPa — visible spurious **bilayer artifact** with a blue "valley" cutting through an otherwise red field (the homogeneous target should be uniform); (b) 6 fields, SM $\in [3.0014, 3.0099]$ MPa — wavy spurious bilayer pattern, red top over blue bottom; (c) 9 fields, SM $\in [3.0001, 3.0001]$ MPa — uniform blue, near perfect. The 9-field case recovers cleanly while 3-field and 6-field hallucinate structure that isn't there.

Implicit (right column): (d) 3 fields, SM $\in [2.7156, 3.0220]$ MPa — mostly uniform red with a small localized blue blob; (e) 6 fields, SM $\in [3.0027, 3.0060]$ MPa — visible vertical gradient/striping; (f) 9 fields, SM $\in [2.9999, 3.0002]$ MPa — almost uniform with faint artifact pattern.

Takeaway: under 5 % noise on a homogeneous target, the explicit method's 3-field and 6-field reconstructions show *spurious bilayer patterns* that don't exist in the ground truth. This is a failure mode worth flagging: the method can invent structure under noise when very few displacement datasets are available. Nine fields restores clean recovery.

Fig. 14. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 1 (5 % noise is introduced into the measured data, Unit: MPa).

[Figure 15 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on **Sample 2** (the two-layer target: background $\mu^0 \approx 3\,\mathrm{MPa}$, top layer $\mu^1 \approx 10\,\mathrm{MPa}$), with **5 % noise** added, using 3, 6, and 9 displacement datasets.

Explicit (left column): bilayer structure visible in all three panels. Colorbar ranges: (a) 3 fields, SM $\in [1.00, 8.91]\,\mathrm{MPa}$ — notably, the colorbar max is **11 % below the nominal 10 MPa** top layer. This is an *undershoot*, opposite to the usual overshoot failure mode. The top layer is not fully recovered with only 3 fields under 5 % noise; (b) 6 fields, SM $\in [1.00, 10.87]\,\mathrm{MPa}$ — slight overshoot, wavy interface; (c) 9 fields, SM $\in [1.00, 10.16]\,\mathrm{MPa}$ — near-nominal with slight overshoot.

Implicit (right column): (d) 3 fields, SM $\in [1.00, 14.20]\,\mathrm{MPa}$ — **42 % overshoot** above the 10 MPa nominal, visible with significant green/yellow patches intruding into the top layer; (e) 6 fields, SM $\in [1.00, 11.42]\,\mathrm{MPa}$; (f) 9 fields, SM $\in [1.00, 11.43]\,\mathrm{MPa}$ — surprising that 9 fields is not clearly better than 6 fields for the implicit method on this sample.

Takeaway: the 3-field explicit case *undershoots* while the 3-field implicit case *overshoots* by 42 % — the two methods fail in opposite directions under low-data + 5 % noise. More displacement fields close the gap for explicit but not cleanly for implicit.

Fig. 15. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 2 (5 % noise is introduced into the measured data, Unit: MPa).

[Figure 16 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on **Sample 3** (the three-layer target: blue bottom $\mu^0 \approx 3\,\mathrm{MPa}$, green middle $\mu^1 \approx 6\,\mathrm{MPa}$, red top $\mu^2 \approx 10\,\mathrm{MPa}$), with **5 % noise** added, using 3, 6, and 9 displacement datasets.

Explicit (left column): (a) 3 fields, SM $\in [1.00, 8.46]$ MPa — undershoots the 10 MPa nominal top layer, matching the 3-field undershoot pattern seen in Fig. 15 for Sample 2; (b) 6 fields, SM $\in [1.00, 10.28]$ MPa; (c) 9 fields, SM $\in [1.00, 10.34]$ MPa. All three panels show wavy but distinguishable three-layer structure.

Implicit (right column): (d) 3 fields, SM $\in [1.00, 15.49]$ MPa — 55 % overshoot, significant green/yellow patches corrupting the top red layer; (e) 6 fields, SM $\in [1.00, 15.98]$ MPa — **worse than 3 fields** on the max-SM axis, a *non-monotone* degradation that the paper's prose does not call out; (f) 9 fields, SM $\in [1.00, 10.17]$ MPa — snaps back close to nominal, cleaner interfaces.

Takeaway: the implicit method's colorbar max on Sample 3 under 5 % noise is *not monotone* in the number of displacement datasets ($15.49 \to 15.98 \to 10.17$ across 3, 6, 9 fields). This is unusual in the paper's overall story; worth noting as a caveat to any "more data always helps" interpretation.

Fig. 16. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 3 (5 % noise is introduced into the measured data, Unit: MPa).

Table 2

The values of the regularization factors utilized in Fig. 8, Fig. 9, Fig. 10, Fig. 11, Fig. 12, Fig. 13, Fig. 14, Fig. 15, Fig. 16.

Inverse method		Explicit inverse method			Implicit inverse method		
Number of surface displacement datasets		3	6	9	3	6	9
Sample 1	no noise	0.00	0.00	5×10^{-27}	2×10^{-14}	5×10^{-14}	1×10^{-13}
	1 % noise	1×10^{-11}	1×10^{-11}	1×10^{-18}	2×10^{-11}	6×10^{-11}	6×10^{-10}
	5 % noise	5×10^{-12}	1×10^{-10}	1×10^{-10}	1×10^{-10}	6×10^{-10}	6×10^{-9}
Sample 2	no noise	0.00	0.00	0.00	1×10^{-14}	1×10^{-14}	1×10^{-14}
	1 % noise	1×10^{-13}	1×10^{-13}	1×10^{-13}	1×10^{-12}	1×10^{-11}	1×10^{-11}
	5 % noise	1×10^{-10}	1×10^{-10}	1×10^{-10}	1×10^{-11}	1×10^{-10}	1×10^{-10}
Sample 3	no noise	0.00	0.00	0.00	1×10^{-14}	1×10^{-14}	1×10^{-14}
	1 % noise	0.00	0.00	0.00	1×10^{-12}	1×10^{-12}	1×10^{-11}
	5 % noise	1×10^{-11}	1×10^{-11}	1×10^{-11}	1×10^{-11}	1×10^{-11}	1×10^{-10}

[Figure 17 — placeholder]

Target shear modulus distribution for the **Case 2** bilayer ring structure: a circular ring with an outer thicker red layer ($\mu_{\text{outer}} = 0.386\,\text{MPa}$) surrounding an inner thinner blue layer ($\mu_{\text{inner}} = 0.129\,\text{MPa}$). Black arrows point radially inward around the perimeter, labeled "displacement measurement", indicating that the inverse problem uses only the displacement measured on the inner wall. Colorbar ticks at $\{0.129, 0.200, 0.267, 0.333, 0.386\}$ MPa (jet colormap, matching the values [37] for physiological soft tissue from the paper's bibliography). The stiffness ratio outer/inner ≈ 3.0 is realistic for arterial wall tissue.

Fig. 17. Target shear modulus distribution of a bilayer ring structure (Unit: MPa).

[Figure 18 — placeholder]

Six-panel optimization trajectory for the **Case 2 bilayer ring structure** (target: outer 0.386 MPa, inner 0.129 MPa) using the explicit inverse scheme with 3 surface displacement datasets (no noise), starting from a 4-layer initial guess. Each panel is a circular ring with a jet colorbar.

(a) Initial guess	4-layer ring, SM $\in [0.01, 0.04]$ MPa
(b) Iteration 20	SM $\in [0.145, 0.438]$ MPa (overshoots outer layer)
(c) Iteration 50	SM $\in [0.131, 0.397]$ MPa, wavy interface emerging
(d) Iteration 100	SM $\in [0.129, 0.393]$ MPa
(e) Iteration 1000	SM $\in [0.131, 0.387]$ MPa (near target)
(f) Iteration 1322 (final)	SM $\in [0.131, 0.384]$ MPa

The outer-layer colorbar max converges to 0.386 MPa *from above*: the trajectory overshoots early ($0.438 \to 0.397 \to 0.393 \to 0.387 \to 0.384$) and settles slightly under the nominal 0.386 target. The ring interface becomes increasingly wavy as iteration proceeds but the overall bilayer structure is recovered faithfully.

Fig. 18. The optimization process with four layers initially utilizing the explicit inverse scheme with 3 surface datasets (Without noise, Unit: MPa).

[Figure 19 — placeholder]

Six-panel comparison of the **Explicit Inverse Method** (top row) vs. **Implicit Inverse Method** (bottom row) on the bilayer ring structure (**no noise**), with 3, 6, and 9 surface displacement datasets.

Explicit (top row): all three reconstructions are visually near-identical to the Fig. 17 target — clean bilayer ring with wavy inner interface. Colorbar ranges: (a) 3 fields, SM $\in [0.131, 0.384]$ MPa; (b) 6 fields, SM $\in [0.131, 0.384]$ MPa; (c) 9 fields, SM $\in [0.131, 0.384]$ MPa. The explicit method reaches the same quality with 3 fields as with 9 — consistent with its low parameter count (56 variables per the paper's earlier claim).

Implicit (bottom row): reconstructions are visibly **dominated by green/yellow artifacts** with no clear bilayer structure recovered. Colorbar ranges show a *monotonically increasing max* as more datasets are added: (d) 3 fields, SM $\in [0.010, 0.478]$ MPa; (e) 6 fields, SM $\in [0.010, 0.632]$ MPa; (f) 9 fields, SM $\in [0.010, 0.813]$ MPa. More data makes the implicit result *worse*, not better — the 9-field max overshoots the 0.386 nominal outer-layer value by **110 %**.

Takeaway: this is the most dramatic failure mode of the implicit method in the entire paper. On the ring structure, with 4030 optimization variables to the explicit method's 56, the implicit method cannot find the bilayer structure at all — increasing displacement data only makes it diverge more. This suggests the implicit method's high dimensionality is *actively harmful* on this problem, not just inefficient.

Fig. 19. The reconstructed results with varying 3, 6 and 9 surface displacement datasets (Without noise, Unit: MPa).

Case 3: A composite with inclusions embedded

In the last example, we replicated a scenario where one or two tumors were embedded within soft tissues with a stiffness ratio of 5:1, as illustrated in Fig. 21. To simulate the forward problem, we applied an "indentation" force on the top surface of a square domain, discretized using 7200 triangular elements. The center point was fixed and other points where on the bottom edge are constrained in the vertical direction, and we only utilized the displacements on the top surface to tackle the inverse problem. To initiate the inverse problem, we employed four inclusions as the initial guess (refer to Fig. 22(a)). These inclusions were predefined in the problem

[Figure 20 — placeholder]

Six-panel comparison of **Explicit Inverse Method** (top row) vs. **Implicit Inverse Method** (bottom row) on the **Case 2 bilayer ring structure** with **1 % noise** added, using 3, 6, and 9 surface displacement datasets.

Explicit (top row): near-identical bilayer ring reconstructions, visually indistinguishable from the Fig. 17 target. Colorbar ranges: (a) 3 fields, SM $\in [0.129, 0.385]$ MPa; (b) 6 fields, SM $\in [0.132, 0.384]$ MPa; (c) 9 fields, SM $\in [0.130, 0.386]$ MPa. The explicit method tolerates 1 % noise essentially perfectly on the ring problem, with all three panels landing within ± 0.002 MPa of the 0.386 nominal outer layer.

Implicit (bottom row): severe overshoot pattern. Colorbar ranges: (d) 3 fields, SM $\in [0.010, 1.251]$ MPa — **224 % overshoot** above the 0.386 nominal outer layer; visually dominated by green/yellow artifacts; (e) 6 fields, SM $\in [0.010, 0.874]$ MPa — 126 % overshoot; (f) 9 fields, SM $\in [0.010, 0.975]$ MPa — 152 % overshoot. The implicit method's max is also non-monotone in this figure ($1.251 \to 0.874 \to 0.975$), continuing the pattern from Fig. 16 and Fig. 19.

Takeaway: as in the noise-free case (Fig. 19), the implicit method is fundamentally failing on the ring problem. The 1 % noise addition does not change the qualitative picture — it is already broken without noise. See Table 3 for the quantitative confirmation (implicit ≈ 41 % vs. explicit ≈ 5 % average relative error).

Fig. 20. The reconstructed results with varying 3, 6 and 9 surface displacement datasets (1 % noise is introduced, Unit: MPa).

Table 3

Comparison of the average relative error in the shear modulus distribution obtained by implicit inverse and explicit inverse methods for the ring structure (1 % noise is introduced).

Inverse method	Explicit inverse method			Implicit inverse method		
Number of surface displacement datasets	3	6	9	3	6	9
Average relative error	4.73 %	5.38 %	4.56 %	42.41 %	40.99 %	41.23 %

Table 4

The values of regularization factors used in Figs. 19 and 20.

Inverse method		Explicit inverse method			Implicit inverse method		
Number of surface displacement datasets		3	6	9	3	6	9
Ring structure	no noise	0.00	0.00	0.00	1×10^{-12}	1×10^{-12}	1×10^{-12}
	1 % noise	0.00	0.00	0.00	1×10^{-12}	1×10^{-12}	1×10^{-12}

[Figure 21 — placeholder]

Target shear modulus distribution for the **Case 3 composite with embedded inclusions**. Two sub-panels:

(a) Sample 1: a square blue background (soft tissue, $\mu_{\text{background}} = 1.00\,\text{MPa}$) with a single red circular inclusion at the center (tumor, $\mu_{\text{inclusion}} = 5.00\,\text{MPa}$). Stiffness ratio 5:1, matching the Case 3 setup described on page 140.

(b) Sample 2: same square background with **two** red circular inclusions side-by-side in the upper half of the domain.

Both panels share boundary conditions: the top edge has an array of small downward-pointing arrows labeled "displacement measurement" (these are the indentation-force application points and where surface displacements are recorded for the inverse problem); the bottom edge is marked with small circles/triangles indicating fixed/vertical-direction constraints. Colorbar ticks at $\{1.00, 2.00, 3.00, 4.00, 5.00\}$ MPa, jet colormap.

Fig. 21. Target shear modulus distribution of composite with inclusions embedded (Unit: MPa).

[Figure 22 — placeholder]

Six-panel optimization trajectory for the **Case 3 Sample 1** (one embedded inclusion, target SM $\approx 5\,\mathrm{MPa}$ against a $1\,\mathrm{MPa}$ soft-tissue background), using the explicit inverse scheme with 3 surface displacement datasets (no noise). The initial guess has **four** inclusions predefined at the four corners of the square domain; the optimization merges them into a single central inclusion as it converges.

(a) Initial guess	4 green circles at corners, SM $\in [1.00, 1.00]\,\mathrm{MPa}$ (uniform)
(b) Iteration 5	4 merging blobs, SM $\in [1.00, 3.20]\,\mathrm{MPa}$
(c) Iteration 10	3 blobs (center two merging), SM $\in [1.00, 3.23]\,\mathrm{MPa}$
(d) Iteration 15	1–2 blobs (red central), SM $\in [1.00, 4.26]\,\mathrm{MPa}$
(e) Iteration 275	1 red inclusion, SM $\in [2.10, 5.32]\,\mathrm{MPa}$
(f) Iteration 328 (final)	1 red inclusion, SM $\in [1.00, 5.10]\,\mathrm{MPa}$

Takeaway: the 4-inclusion initialization does NOT cause the algorithm to find 4 tumors. Instead, the four components merge into the single ground-truth tumor by iteration 15. This is the opposite of the usual over-parameterization concern: redundant initial components collapse to the correct solution rather than forcing a spurious multi-component reconstruction. The explicit parameterization has 69 total variables (4 inclusions $\times$ 16 geometric parameters + 4 shear moduli + 1 background shear modulus).

Fig. 22. The optimization process with four inclusions initially utilizing the explicit inverse scheme for a solid with inclusion embedded when 3 surface displacement datasets are utilized (Without noise, Unit: MPa).

domain. Each inclusion was represented by 16 geometric parameters and 1 material property, which is the shear modulus. Consequently, the total number of optimization variables reached 69, inclusive of the shear modulus value of the background.

During the optimization process, the four predefined components gradually merged into one single inclusion. As convergence was reached, the shear modulus of this inclusion approached the target values effectively, demonstrating the success of the inverse problem-solving method. The reconstructed results, as shown in Fig. 22, illustrated the accurate estimation of the shear modulus for the inclusion after the convergence of the algorithm. Indeed, the explicit inverse method demonstrated a notably faster convergence rate compared to the implicit inverse method. Specifically, the explicit inverse method achieved convergence in just 328 iterations,

[Figure 23 — placeholder]

Six-panel comparison of the **Explicit Inverse Method** (top row) vs. **Implicit Inverse Method** (bottom row) on **Case 3 Sample 1** (one embedded inclusion, target $\approx 5\,\mathrm{MPa}$), with **no noise**, using 3, 6, and 9 surface displacement datasets.

Explicit (top row): (a) 3 fields, SM $\in [1.00, 5.10]\,\mathrm{MPa}$; (b) 6 fields, SM $\in [1.00, 5.06]\,\mathrm{MPa}$; (c) 9 fields, SM $\in [1.00, 4.97]\,\mathrm{MPa}$. All three panels recover a clean single red inclusion on a blue background, near the 5 MPa nominal target. The colorbar max drifts slightly downward as more fields are added ($5.10 \to 5.06 \to 4.97$).

Implicit (bottom row): (d) 3 fields, SM $\in [1.00, 4.47]\,\mathrm{MPa}$; (e) 6 fields, SM $\in [1.00, 4.58]\,\mathrm{MPa}$; (f) 9 fields, SM $\in [1.00, 4.79]\,\mathrm{MPa}$. **All three panels undershoot the 5 MPa nominal target**, with the 3-field case falling short by 10.6 %. The max values are non-monotone in dataset count ($4.47 \to 4.58 \to 4.79$) but monotonically approach the target as more fields are added — the opposite direction from the explicit method's slight drift.

Takeaway: on the one-inclusion case without noise, both methods produce visually similar reconstructions, but the explicit method hits the target nominal from above (slight overshoot) while the implicit method hits it from below (persistent undershoot). More data helps the implicit method approach nominal but doesn't quite reach it.

Fig. 23. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 1 (Without noise, Unit: MPa).

while the implicit inverse method required a significantly higher number of iterations, reaching convergence after 4354 iterations.

In the absence of noise in the datasets, both the explicit and implicit inverse methods exhibited excellent performance in yielding shear modulus reconstructions, as demonstrated in Fig. 23. However, when 1 % noise was introduced to the datasets, the explicit inverse method outperformed the implicit inverse method (see Fig. 24). For the explicit inverse method, the average relative error in the reconstructed shear modulus values was approximately 2 %, indicating a relatively accurate estimation despite the presence of noise (see Table 5). On the other hand, the implicit inverse method exhibited a higher average error of more than 3 %, signifying a less precise reconstruction in the presence of noise. Overall, the explicit inverse method proved to be more robust and capable of handling noisy datasets, leading to more accurate shear modulus reconstructions compared to the implicit inverse method. Furthermore, it is noteworthy that the reconstructed results maintain good quality even when the noise level is increased to 5 % (see Fig. 25).

In the case where two inclusions were embedded in the background (as depicted in Fig. 21(b)). As shown in Fig. 26, the results obtained using the implicit inverse method showed a consistent underestimation of the recovered shear modulus values by at least 20 % as the size of the mapped inclusions increased. Additionally, the shape of the mapped inclusions appeared to become more circular with the increase of surface displacement datasets. Conversely, when employing the explicit inverse method, the shear modulus values within the two inclusions were overestimated by approximately 20 %. Furthermore, the shape of the recovered inclusions exhibited a more circular appearance compared to those predicted by the implicit inverse method. The regularization factors used in Figs. 23–26 are listed in Table 6.

To assess the robustness of the proposed explicit inverse method, we varied the shape of the inclusion in Sample 1, as depicted in Fig. 27(a) and (d). Our observations indicate that the reconstructed results obtained from the explicit inverse method effectively preserve the shape of the complex inclusion, even with only three displacement datasets. In contrast, the implicit inverse method failed to achieve this level of accuracy.

Discussion

The paper proposes an explicit inverse approach for characterizing the nonhomogeneous elastic property distribution of three typical nonlinear biological structures with surface deformation. This approach is different from the prevalent implicit inverse approach in that it optimizes only the geometric parameters and shear moduli of each region, reducing the scale of the optimization problem. By doing so, the proposed method improves the uniqueness of the inverse solution since the total number of optimization variables is reduced. Additionally, the paper highlights an important advantage of the explicit method: it only requires surface deformation data, which can be measured with high accuracy using low-cost measurement techniques. This makes the proposed approach more practical and accessible for real-world applications, as it avoids the need for complex and expensive instrumentation to gather the full-field data.

[Figure 24 — placeholder]

Six-panel comparison of **Explicit Inverse Method** (top row) vs. **Implicit Inverse Method** (bottom row) on **Case 3 Sample 1** (one embedded inclusion, target $\approx 5\,\mathrm{MPa}$), with **1 % noise** added, using 3, 6, and 9 surface displacement datasets.

Explicit (top row): (a) 3 fields, SM $\in [1.00, 5.75]\,\mathrm{MPa}$ — **15 % overshoot** of the 5 MPa nominal tumor stiffness; (b) 6 fields, SM $\in [1.00, 5.30]\,\mathrm{MPa}$; (c) 9 fields, SM $\in [1.00, 5.01]\,\mathrm{MPa}$ — near-perfect with 9 fields. The max drifts downward monotonically ($5.75 \to 5.30 \to 5.01$) toward the nominal target.

Implicit (bottom row): (d) 3 fields, SM $\in [1.00, 3.26]\,\mathrm{MPa}$ — **35 % undershoot**, a dramatically worse number than the paper's prose claim of "$\approx 3\,\%$ average error" suggests (the prose error is measured over the whole domain, not the inclusion max); (e) 6 fields, SM $\in [1.00, 3.51]\,\mathrm{MPa}$ — 30 % undershoot; (f) 9 fields, SM $\in [1.00, 4.82]\,\mathrm{MPa}$ — still 3.6 % below nominal. Max drifts upward monotonically ($3.26 \to 3.51 \to 4.82$) toward the target but doesn't reach it.

Takeaway: with 1 % noise, the explicit and implicit methods fail in *opposite directions* on this problem — explicit *overshoots* the tumor stiffness, implicit *undershoots* it. Both approach the 5 MPa target from their respective sides as more fields are added, but the explicit method gets closer faster. The prose's "2 % vs 3 % average error" framing understates the practical significance: a 35 % undershoot of the tumor stiffness (panel d) would have very different clinical implications than a 2 % average-domain error suggests.

Fig. 24. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 1 (1 % noise is introduced into the measured data, Unit: MPa).

Table 5

Comparison of the average relative error in the shear modulus distribution obtained by implicit inverse and explicit inverse methods for the case with one inclusion embedded (Figs. 24 and 25).

Inverse method	Explicit inverse method			Implicit inverse method		
Number of surface displacement datasets	3	6	9	3	6	9
1 % noise	1.81 %	1.50 %	1.60 %	4.28 %	3.58 %	3.41 %
5 % noise	1.88 %	1.86 %	1.74 %	6.49 %	6.05 %	4.26 %

[Figure 25 — placeholder]

Six-panel comparison of **Explicit Inverse Method** (top row) vs. **Implicit Inverse Method** (bottom row) on Sample 1 with **5 % noise** added, using 3, 6, and 9 displacement datasets.

Explicit (top row): (a) 3 fields, SM $\in [1.00, 5.83]$ MPa — 17 % overshoot; (b) 6 fields, SM $\in [1.00, 6.09]$ MPa — **22 % overshoot, WORSE than 3-field**; (c) 9 fields, SM $\in [1.00, 4.66]$ MPa — now 7 % *undershoot*. The max is non-monotone ($5.83 \to 6.09 \to 4.66$) and crosses the nominal 5 MPa target between 6 and 9 fields. This contradicts the intuitive "more data $\Rightarrow$ better" story.

Implicit (bottom row): (d) 3 fields, SM $\in [1.00, 2.90]$ MPa — **42 % undershoot**; (e) 6 fields, SM $\in [1.00, 3.21]$ MPa; (f) 9 fields, SM $\in [1.00, 3.40]$ MPa — still 32 % undershoot. The implicit method never approaches the 5 MPa target under 5 % noise; even 9 fields leaves it severely below nominal.

Table 5 vs. this figure — an important disconnect: Table 5 reports the explicit-9-fields-5%-noise case as *1.74 % average relative error*, but panel (c) shows the tumor-stiffness max at *4.66 instead of 5.00 MPa*. Both numbers are correct — Table 5 averages over the whole 2D domain (which is mostly 1 MPa background where any method is trivially correct), while the figure shows the peak inside the tumor itself. A clinician looking at the "≈ 2 % error" claim would expect a ± 0.1 MPa estimate of the tumor; the figure shows a -7 % bias in the quantity that actually matters. The prose should have reported tumor-specific error alongside the domain-average metric.

Fig. 25. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 1 (5 % noise is introduced into the measured data, Unit: MPa).

In the numerical cases presented in the paper, the total number of predefined interfaces (for layered structures) or components (for composites with embedded inclusions) in the initial guesses is intentionally kept not much larger than the actual target. This is because the topological information of the elastic property distribution can often be determined beforehand based on prior knowledge or observations. For example, in biomedical imaging, characteristics such as the number, location, and rough shape of tumors can be observed and identified. Similarly, for typical layered biological tissues like skins or corneas, the total number of layers has likely been well studied through in vitro experiments or other research. Due to this available prior knowledge, it is unnecessary to set a large number of deformable components or interfaces in the initial guess for the explicit inverse method. By constraining the initial guesses to a reasonable and relevant number of interfaces or inclusions, the optimization process can be more focused and efficient. This can lead to faster convergence and more accurate characterization of the elastic property distribution with a reduced risk of convergence to a non-physical solution.

However, it should be noted that the results of this study do not conclusively demonstrate that the explicit inverse method outperforms the implicit inverse method in all cases. In scenarios where the elastic property distribution lacks significant heterogeneity or where no prior knowledge is available, the explicit inverse method may indeed perform less effectively compared to the implicit inverse method. This is because the explicit inverse method relies on predefined interfaces or inclusions, which may not accurately capture the complexity of the underlying distribution. In such situations, the implicit inverse method, with its greater flexibility and ability to explore a wider range of possibilities, might yield more accurate and robust results. The choice between the explicit and implicit inverse methods, therefore, depends on the specific characteristics of the problem and the availability of prior information, ensuring the selection of the most suitable approach for achieving meaningful and reliable results.

Due to the limited availability of experimental datasets, this paper primarily focuses on numerical verification. It should be noted that the implicit inverse method has demonstrated successful applications in reconstructing the nonhomogeneous shear modulus distribution of soft solids using experimental datasets [38]. However, considering the superior performance of the explicit inverse solver in numerical validation, we are confident that it can achieve even better results when applied to the experimental datasets. In

[Figure 26 — placeholder]

Six-panel comparison of **Explicit Inverse Method** (top row) vs. **Implicit Inverse Method** (bottom row) on **Case 3 Sample 2** (two embedded inclusions side-by-side, target ≈ 5 MPa each), with **no noise**, using 3, 6, and 9 displacement datasets.

Explicit (top row) — both inclusions are recovered as distinct red blobs on a uniform blue background. Colorbar ranges: (a) 3 fields, SM $\in [1.00, 7.06]$ MPa — **41 % overshoot**; (b) 6 fields, SM $\in [1.00, 6.06]$ MPa — 21 % overshoot; (c) 9 fields, SM $\in [1.00, 5.97]$ MPa — 19 % overshoot. Max drifts downward monotonically as more fields are added, approaching but not reaching the 5 MPa nominal.

Implicit (bottom row) — both inclusions are recovered as green/yellow blobs (distinctly underbright relative to the explicit method's red). Colorbar ranges: (d) 3 fields, SM $\in [1.00, 2.61]$ MPa — **48 % undershoot**; (e) 6 fields, SM $\in [1.00, 3.00]$ MPa — 40 % undershoot; (f) 9 fields, SM $\in [1.00, 3.82]$ MPa — 24 % undershoot. Max drifts upward monotonically, approaching but nowhere close to the 5 MPa nominal.

Takeaway: on the two-inclusion case (even without noise), the explicit method overshoots by 19–41 % and the implicit method undershoots by 24–48 %. The paper's prose reports both deviations (see page 18: "overestimated by approximately 20 %" for explicit, "underestimation ... by at least 20 %" for implicit). In this case the prose framing and the visible data agree — unlike the Fig. 25 case where the "2 % average error" claim hid a 7 % inclusion-specific undershoot.

Fig. 26. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 2 (Without noise, Unit: MPa).

Table 6

The regularization factors of Figs. 23–26.

Inverse method		Explicit inverse method			Implicit inverse method		
Number of surface displacement datasets		3	6	9	3	6	9
Sample 1	no noise	1×10^{-15}	1×10^{-16}	1×10^{-17}	1×10^{-14}	1×10^{-14}	1×10^{-14}
	1 % noise	1×10^{-13}	1×10^{-13}	2×10^{-13}	5×10^{-13}	3×10^{-13}	2×10^{-12}
	5 % noise	1×10^{-12}	1×10^{-12}	1×10^{-12}	1×10^{-11}	1×10^{-11}	1×10^{-12}
Sample 2	no noise	0.00	0.00	0.00	1×10^{-13}	1×10^{-13}	1×10^{-14}

future research, we plan to explore the integration of optical measurement techniques with the proposed explicit inverse approach to conduct practical experiments and further test this novel structural health monitoring approach. By combining optical measurement data with our method, we aim to validate its effectiveness in real-world scenarios and enhance its applicability for monitoring the variation of biomechanical behavior of soft tissues. This would provide valuable insights into the method's performance under realistic conditions and potentially open up new avenues for its practical implementation in biomedical applications.

Overall, the explicit inverse approach presented in the paper offers a more efficient and cost-effective method for characterizing the elastic property distribution of biological structures with surface deformation. The reduction in optimization variables and the reliance on easily measurable surface deformation data make it a valuable contribution to the field.

Conclusion

In this paper, we propose an explicit inverse approach to map the nonhomogeneous elastic property distribution of three typical tissue structure by surface deformation nondestructively. Several numerical examples are employed to assess the feasibility and performance of the proposed method. The results of the numerical study demonstrate the robust capability of the explicit inverse method in accurately reconstructing the nonhomogeneous shear modulus distribution of soft tissues, surpassing the performance of the implicit inverse method with limited surface deformation datasets. Specifically for the layered ring structure, we observe that the average relative error of the reconstruction can exceed 40 % when using the implicit inverse method, while the explicit inverse method achieves a remarkable average relative error of only 5 %. The outcomes of our research underscore the efficacy of the proposed explicit inverse approach, offering a more efficient and cost-effective means to characterize the elastic property distribution of biological structures utilizing a reduced number of optimization variables.

[Figure 27 — placeholder]

Six-panel comparison of **two different inclusion shapes** reconstructed with 3 displacement datasets (Sample 1, no noise). The two rows show different target shapes; the columns show target, explicit method result, and implicit method result.

Top row (irregular/elongated inclusion): (a) target: red elongated/irregular shape, SM $\in [1.00, 5.00]$ MPa; (b) Explicit, 3 fields: SM $\in [1.00, 5.00]$ MPa — the reconstructed shape faithfully preserves the irregular target geometry; (c) Implicit, 3 fields: SM $\in [1.00, 4.98]$ MPa — the shape is *distorted*, rendered as a green/yellow blob without the sharp edges of the target, and fails to capture the elongated profile.

Bottom row (square inclusion): (d) target: red square shape, SM $\in [1.00, 5.00]$ MPa; (e) Explicit, 3 fields: SM $\in [1.00, 5.00]$ MPa — the reconstructed shape is a rounded-square, closely matching the target; (f) Implicit, 3 fields: SM $\in [1.00, 4.71]$ MPa — the shape is rendered as a green amorphous blob, significantly distorting the square target.

Takeaway: this figure is the qualitative case for the explicit inverse method's main selling point on non-circular inclusion geometries. The explicit method's B-spline-based shape parameterization allows it to faithfully represent elongated or square shapes; the implicit method's pixel-based optimization reverts to blob-like reconstructions regardless of the target shape. This is a shape-fidelity argument, not a shear-modulus accuracy argument — and it's where the explicit method's prior knowledge (topology + rough shape) pays off most visibly.

Fig. 27. The target and reconstructed results for varying shapes of inclusion with 3 displacement datasets corresponding to Sample 1 (Without noise, Unit: MPa).

CRediT authorship contribution statement

Yue Mei: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Dongmei Zhao:** Writing – review & editing, Writing – original draft, Software, Formal analysis, Data curation. **Changjiang Xiao:** Writing – original draft, Investigation, Formal analysis, Data curation. **Zhi Sun:** Methodology, Formal analysis, Data curation. **Weisheng Zhang:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition. **Xu Guo:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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